

Ruthless Controller Relay — What Works Cheat Sheet

Plain-English guide: which controllers work, how well, and why. No tech knowledge needed.

The short version

Best experience: plug your controller in with a USB cable. Every controller does everything (moving, buttons, vibration, and light color) when it's wired. Bluetooth (wireless) works great for moving and buttons on every controller, but PlayStation controllers (DualSense / DualShock 4) lose vibration and light color over Bluetooth — see why below. This is not something wrong with your setup, it's true for everyone, everywhere.

Compatibility matrix

Controller	How it's connected	Movement & Buttons	Vibration (Rumble)	Light Color	Notes
PS5 Controller (DualSense)	USB Cable	Works	Works	Works	Everything works. Recommended.
PS5 Controller (DualSense)	Bluetooth (wireless)	Works	Doesn't work	Doesn't work	Plug in a cable if you want rumble/light.
PS4 Controller (DualShock 4)	USB Cable	Works	Works	Works	Everything works. Recommended.
PS4 Controller (DualShock 4)	Bluetooth (wireless)	Works	Doesn't work	Doesn't work	Plug in a cable if you want rumble/light.
Xbox Controller	USB Cable	Works	Works	n/a *	Fine for normal play. HOTAS mode does NOT reliably work this way (V / M unreliable too) — see warning below (†).
Xbox Controller	Wireless Dongle (official Xbox Wireless Adapter, or the controller's own)	Works	Works	n/a *	Fully works, HOTAS/V/M included. Recommended over a cable for this controller specifically.
HOTAS / Flight Stick / Throttle	USB Cable	Works	n/a *	n/a *	These don't have vibration motors or a color light in real life, so there's nothing to turn on.

* n/a = that controller doesn't physically have this feature, so it's not a bug or limitation of the software.

⚠ + HOTAS mode does NOT reliably work with a wired Xbox controller

HOTAS mode is the main reason this program exists — and a wired Xbox controller is the one case where it can't be trusted. Windows connects a wired Xbox controller through a special driver that this software (or anything else) has no ability to hide from other apps, unlike every other controller in this list. That means the game can still notice the real controller and use normal gamepad controls instead of reading the joystick axes this program is feeding it through HOTAS mode — the mode may look like it's working on this screen while the game itself ignores it. The same underlying problem also makes **V** (disguise as PlayStation) and **M** (button remap) unreliable for this controller.

Fix: connect that Xbox controller through a wireless dongle instead of a cable (the official Xbox Wireless Adapter, or the controller's own 2.4GHz dongle if it has one) — same controller, same buttons, just a different plug — and HOTAS mode, V, and M all work exactly as expected.

Why doesn't PlayStation Bluetooth rumble/light work?

It's not broken — nobody has cracked this. When a PS4/PS5 controller connects over a USB cable, it accepts vibration and light-color commands using a simple, well-understood method. Over Bluetooth, Sony uses a different, much stricter method that basically no software outside of an actual PlayStation console has ever gotten working reliably — including big well-known controller tools that have been trying for years. We tested it thoroughly (including trying a second, different Bluetooth adapter) and confirmed the same result every serious attempt online has run into. Moving and pressing buttons over Bluetooth is completely unaffected — only vibration and the light color are impacted.

Two versions are included — which one do I run?

Version	What it does	When to use it
RuthlessControllerRelay.exe	Full version — vibration, light color, all the extras from the matrix above.	Use this one first. This is the normal, recommended version.
RuthlessControllerRelay_SAFE.exe	Stripped-down version — no vibration, no light color, no battery reading. Just clean, reliable button/stick input.	Only use this as a backup if the full version ever acts strange on your PC. It's a simpler fallback with fewer moving parts.

Controls — what the keys do

Key	What it does
Ctrl + Alt + H (or hold Back + Start on the controller)	Switch between Normal mode (game sees your controller normally) and HOTAS mode (game sees a joystick). Do this once at the start — this is the main control.
V	Flip the OUTPUT badge between PlayStation-style and Xbox-style — use this if a specific game only recognizes one type.
R	Change which key or controller-button-combo does the Normal/HOTAS switch above.
M	Remap a button — make one button act like a different one.
G	Pick which game this launches, or launch it directly next time.

Type these keys into the black program window itself (click it first so it's the active window), not into your game.

Reading the screen: right vs. wrong

The top banner and the two INPUT/OUTPUT boxes are the whole story — glance at those first any time something feels off.

✓ Everything's fine

- Banner reads **NORMAL MODE** or **HOTAS MODE** — colored **teal**, **purple**, or **green** (never orange or red)
- **Controller:** line names your actual controller (e.g. "PlayStation 5 controller (DualSense)")
- Both INPUT and OUTPUT boxes show a real type — **PS**, **XBOX**, or **HOTAS** (never blank dashes)
- Sticks/bars/buttons on screen move the instant you touch the real controller

This is your green light — safe to start the game.

X Something's wrong

- **"NOT READY — waiting for a controller..."** (orange/red banner)
Fix: turn the controller on / plug it in. It's picked up automatically — no restart needed. Don't start your game until this clears.
- **"MODE UNKNOWN — press Ctrl+Alt+H once to set it"** (red banner)
Fix: press Ctrl+Alt+H once to choose Normal or HOTAS mode.
- OUTPUT box shows "----"
Fix: no virtual controller made yet — set the mode (above) first; check the log file if it still doesn't clear.
- "This window is too short to show the dashboard..."
Fix: maximize/enlarge the window, or shrink the font — it fixes itself automatically once there's room.
- Banner turns **red** while playing (battery)
Fix: only happens over Bluetooth — plug in a charging cable, wired play never shows this.

Two more live indicators worth knowing:



- The **face buttons light up their real-life color** when pressed (PlayStation Triangle/Square/Cross/Circle or Xbox Y/X/A/B) — unpressed buttons stay plain gray. If a button never lights up, that specific button isn't registering.
- **Banner color meanings in full:** teal = Normal mode, output matches your real controller type (PS→PS or Xbox→Xbox); purple = Normal mode but output is set to the other type on purpose (e.g. real PS controller showing up as Xbox — this is fine if you did it with V); green = HOTAS mode; orange/red = mode not chosen yet or no controller found, see the table above.

Something not working?

Every run writes a plain-text log file next to the program (**driver_install_log.txt**). If anything looks wrong, that file has the details — send it along if you're asking for help troubleshooting. There's also an **Uninstall.bat** included that cleanly removes everything the program installed, if you ever want to fully undo it.